

Writing a variable

```
let headY = 190;
// \  /   \  /   \  /
// make  the  what goes
// a name name  in it
```

Use one when

- the number has to **change** while the sketch runs
- the same number appears in **several places**

mouseX and mouseY

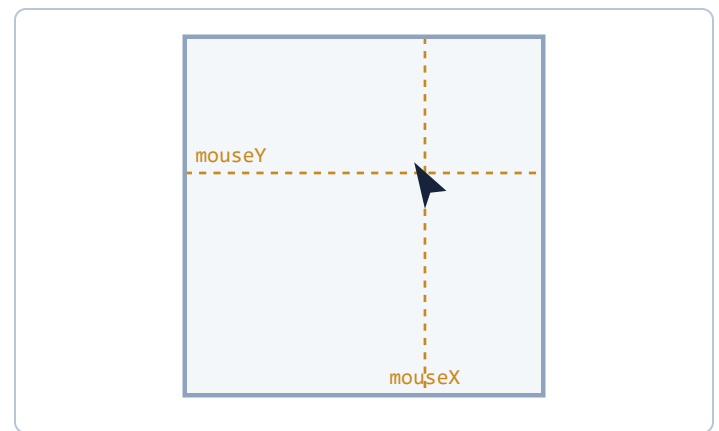
Two variables p5 already made. They hold where the cursor is **right now**, updated before every frame.

```
circle(mouseX, mouseY, 40);
```

You only **read** them. p5 writes them. A variable you made and one p5 made are the same idea — a name for a number — they just have different owners.

Where it goes

- let at the **top**, outside draw()
- changing it goes **inside** draw()

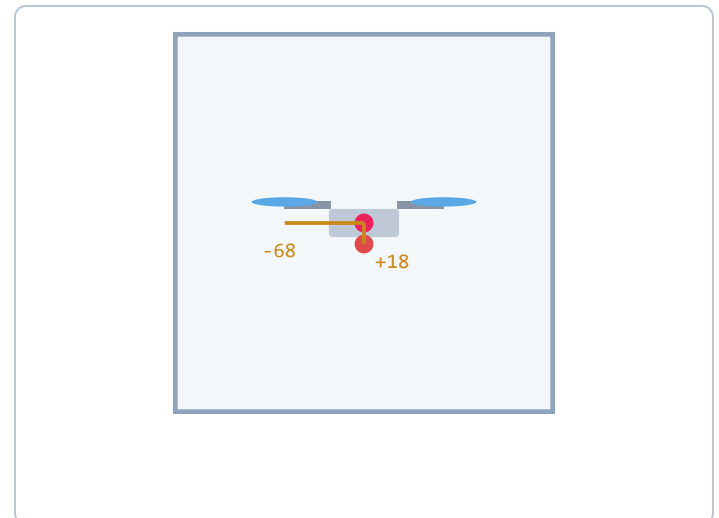


Moving a whole object

Pick **one point**. Write every other part as a distance from it — left or right, up or down.

```
rect(mouseX, mouseY, 60, 24);
rect(mouseX - 48, mouseY - 14, 40, 6);
rect(mouseX + 48, mouseY - 14, 40, 6);
circle(mouseX, mouseY + 18, 16);
```

PART	ACROSS	DOWN
body	mouseX	mouseY
left arm	- 48	- 14
right arm	+ 48	- 14
camera	same	+ 18



Only positions move. Sizes never do. 60, 24, 40, 6 stay exactly as they are. A drone does not get bigger when it flies right — if yours stretches sideways, you converted a size by mistake.

Check it in the corners. Near the middle of the canvas, a part that is twenty pixels off still looks about right. Move the mouse into all four corners — anything left behind still has a fixed number.

On the AP exam

What you write

```
let y = 100;  
y = y + 1;
```

Exam Reference Sheet

```
y ← 100  
y ← y + 1
```

The AP exam is not written in JavaScript. It uses College Board’s own notation, and the arrow `←` means **becomes** — the same word we use for `=`. No `let`, no semicolon.

Where this fits in AP CSP

Big Idea 3 — Algorithms and Programming · 30–35% of the AP Exam

CODE	LEARNING OBJECTIVE
3.1	Variables and Assignments
AAP-1.A	Represent a value with a variable
AAP-1.B	Determine the value of a variable as a result of an assignment
3.A	<i>Skill:</i> generalize data sources through variables

Skill 3.A is what this lesson is really about. A drone written with `mouseX` works anywhere on the canvas. One written with `200` works in exactly one place. That is generalizing.

Same technique, two sources. Yesterday the point was a variable you changed yourself. Today it is one p5 changes for you. Either way: pick one point, and measure everything else from it.